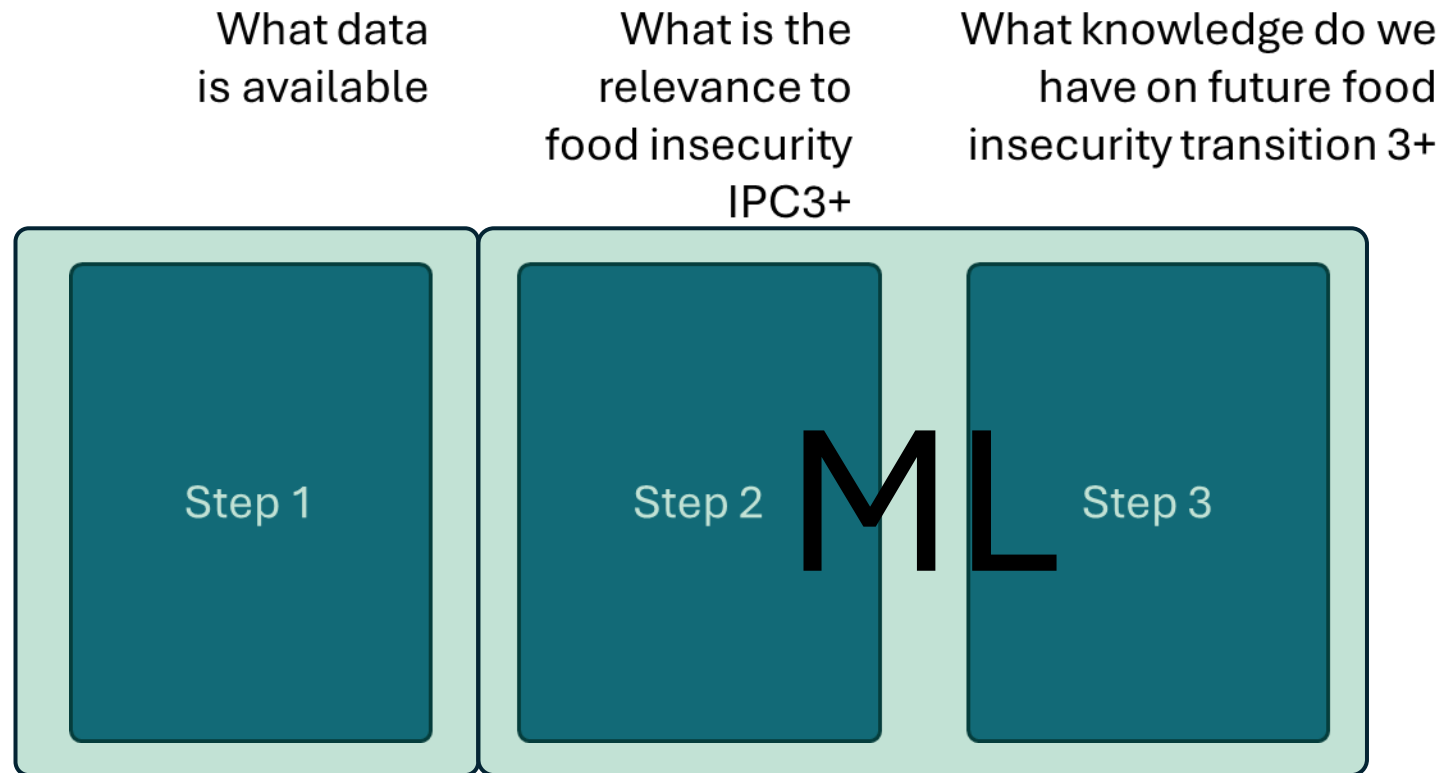


Forecasting Transitions to Acute Food Insecurity to Support Anticipatory Action

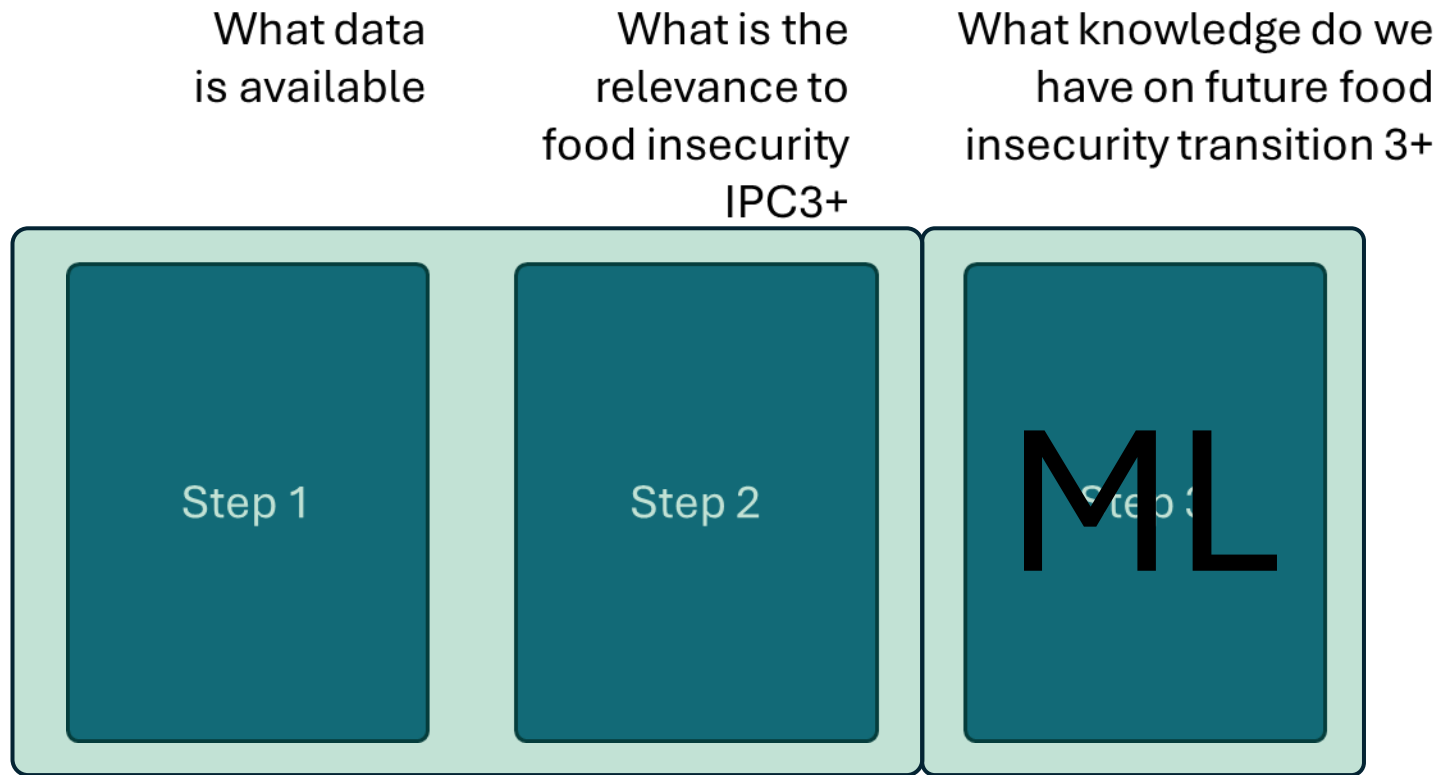
Transferrable Features for Rare Events

- Transitions to acute food insecurity are rare
- Complex interplay of factors: weather, conflict, markets, geopolitics, natural hazards
- Heterogeneity across regions shape vulnerability and resilience
- Heterogeneity across regions complicates generalization of predictive models
- Africa: diverse climates, geographies, economies, ethnic backgrounds, languages, religions and varying political systems
- Rare events require leveraging information across regions and time
- Signals must be comparable across countries despite structural differences
- Goal: transform diverse inputs into common, comparable representations

Big data approach to ML



Transferrable Features for Rare Events



What data
is available

One weather shock

- Crops - geolocated
- Exceedance
Temperature
- Events - geolocated
- Crops calendars -
geolocated
- Sensitivity
(production
volume) of staple
crop species to leaf
burning - temporal
- Sensitivity
(production value)
of cash crop
species to leaf
burning - temporal
- [weather shock
precursors –
ENSO...]

What is the
relevance to
food insecurity
IPC3+

**Impact from weather shocks
of that region – time t-1,t-2,t-3,t-4**

**Price shock of staple calories in
local markets t-1,t-2,t-3,t-4**

- **Regional transmission weather
shocks**
- **Regional transmission conflict
displacement**
- **International price
transmission**
- **International cash crops
transmission**

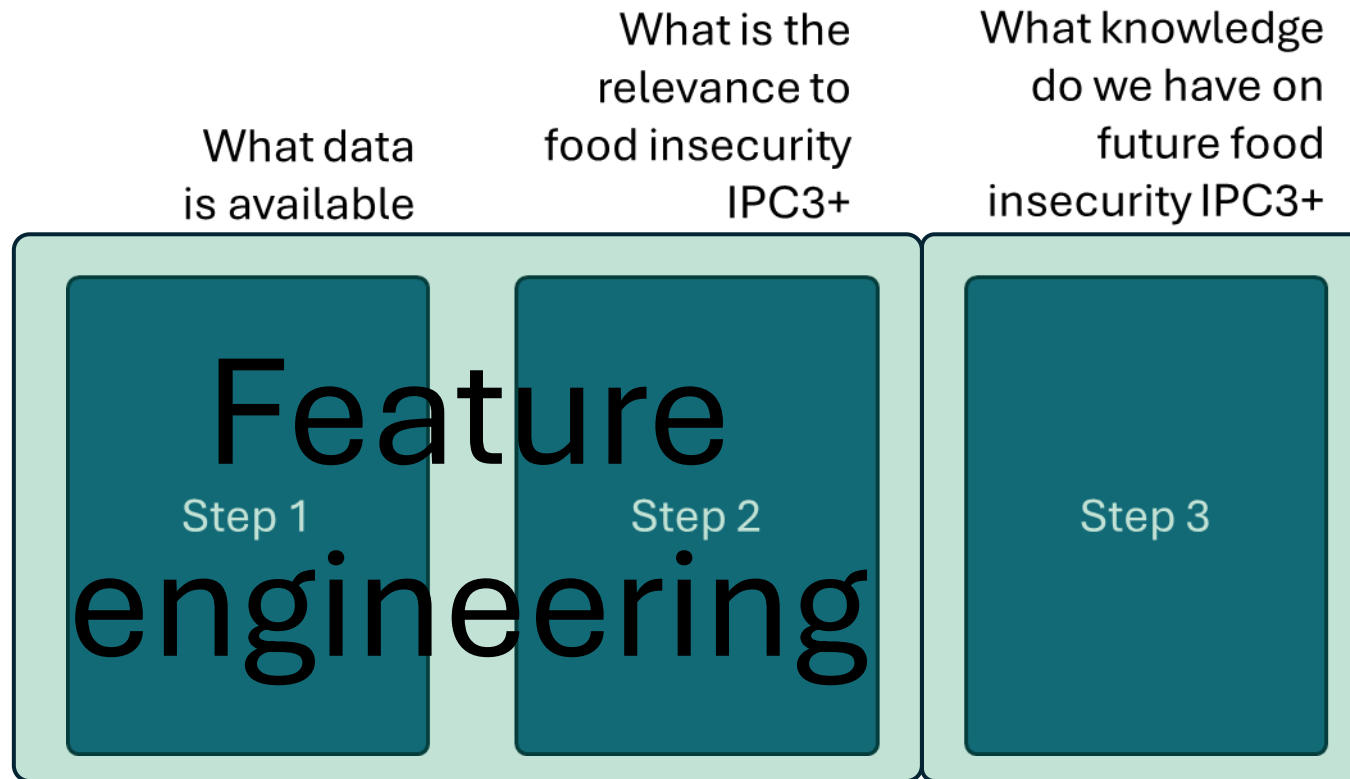
**Energy shocks, trade policy shocks
...**

**Fixed effects – governance, development, that
increase susceptibility to shocks.**

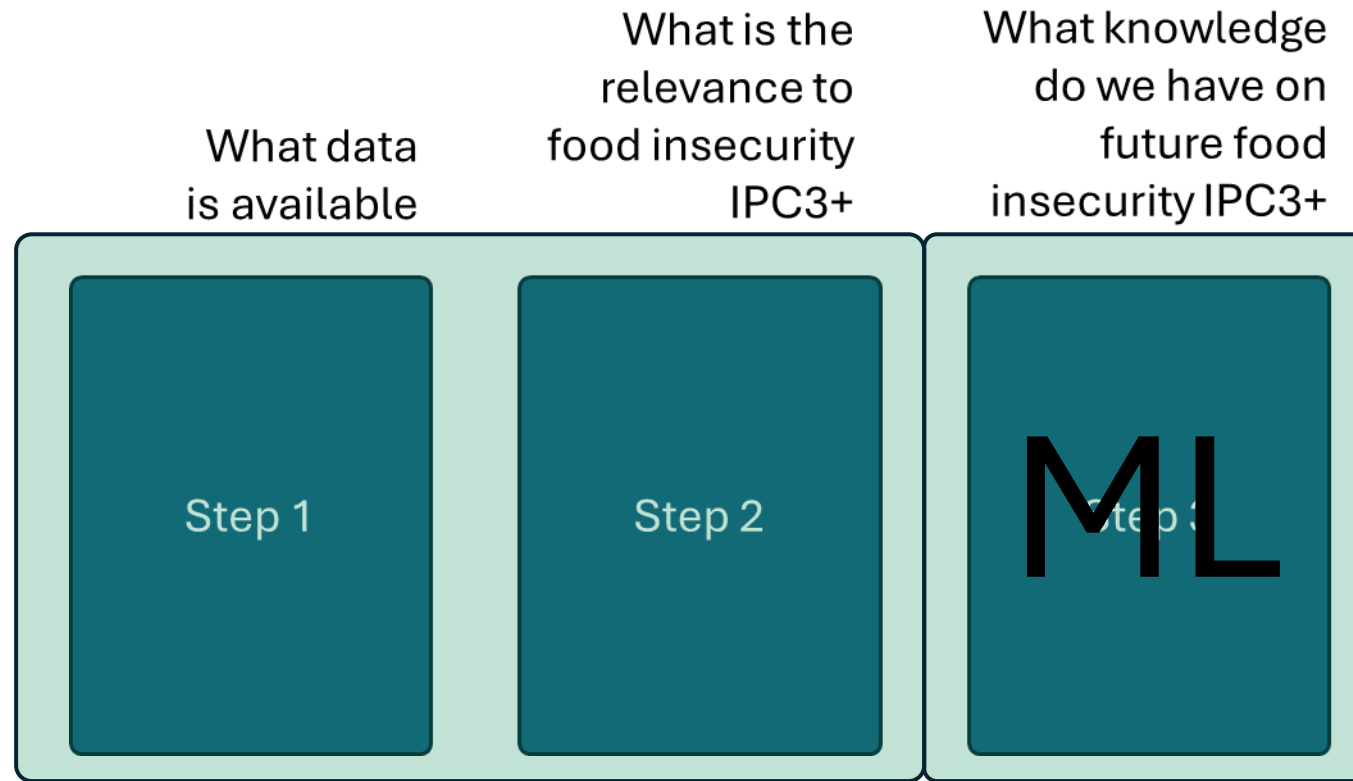
What knowledge
do we have on
future food
insecurity IPC3+

- Will the Admin2
region become
IPC3+ (severe acute
food insecurity) next
quarter?
- How many new
people will be in
severe food
insecurity next
quarter?
- Why did it transit to
sever food
insecurity?
- What observable
parameters getting
worse 3-6 months
ago reliably and
proportionally
indicate acute food
insecurity numbers
getting worse?

Transferrable Features for Rare Events



Modelling

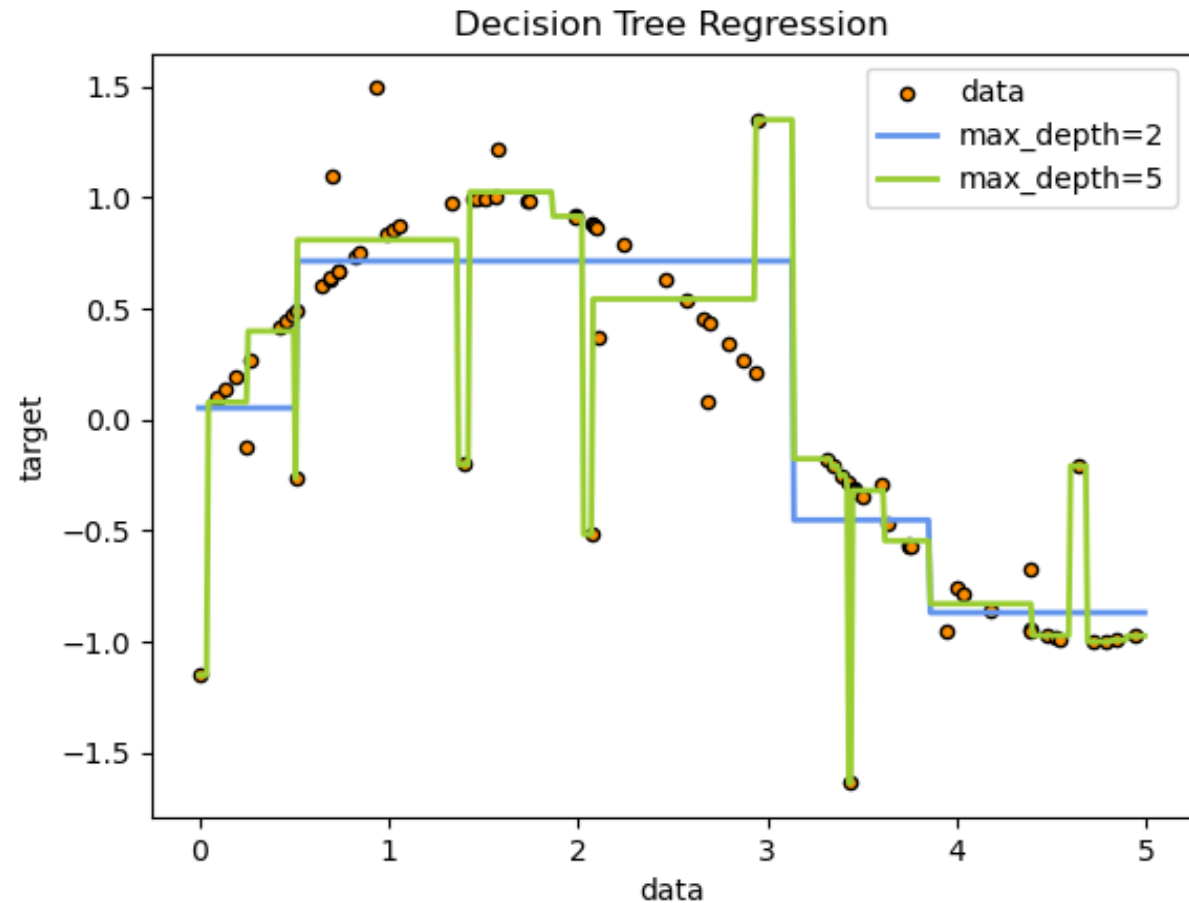


Target Variable: Population in Acute Food Insecurity (IPC 3+)

- FEWS NET food security classifications
- Near Term projections (ML1) and Current Situation (CS) assessments
- Each area is classified on the IPC scale (1–5)
- IPC 3+: acute food insecurity
- We combine these classifications with population data
- For each subnational region (GAUL2):
 - Estimate the share of the population in IPC 3+
 - Final target is a percentage between 0 and 1
- Transform a categorical map into a continuous metric

Modelling for Prediction: Decision Trees

- Maximize predictive accuracy of food insecurity outcomes
- Capture nonlinearities across space and time
- **Gradient-Boosted Decision Trees (XGBoost):**
 - Ensemble of decision trees trained via gradient boosting.
- Why tree-based boosting:
 - Strong empirical performance in forecasting tasks with tabular time series.



Weighted Learning for Rare Transitions

- Since transitions are rare, data thus models will underweight these events
- We assign higher importance (weight) to observations where there are transitions into IPC 3+ conditions
 - Transition to IPC3+ → more weight
 - Same situation as last observation → less weight
- Weighting is controlled by a scaling parameter:
 - Higher scale → more emphasis on transitions
 - Lower scale → more balanced learning across all observations
- Tell the model to pay disproportionate attention to emerging crises

$$w_t = 1 + \text{scale} \cdot \frac{\max(y_t - y_{t-1}, 0)}{\Delta t}$$

Results Overview

- Analysis based on historical data from 2011 to 2025
- Coverage includes multiple countries across Sub-Saharan Africa:
 - *Burundi, Kenya, Madagascar, Malawi, Mozambique, Somalia, South Sudan, Uganda, Zimbabwe, Ethiopia, Chad, Democratic Republic of the Congo, Cameroon, Sudan*
- Results focus on predicting population share IPC 3+
 - At subnational (GAUL2) level
 - Across time (monthly)
- We compare model outputs with FEWS NET projections
 - The current benchmark used to inform anticipatory action decisions
- This is **not a replacement for expert analysis**
- The model as a decision-support tool, the goal is to:
 - Provide a complementary, data-driven signal
 - Help identify emerging risks and anomalies
 - Support analysts in prioritizing attention and resources

An Ensemble Approach to Capture Transitions

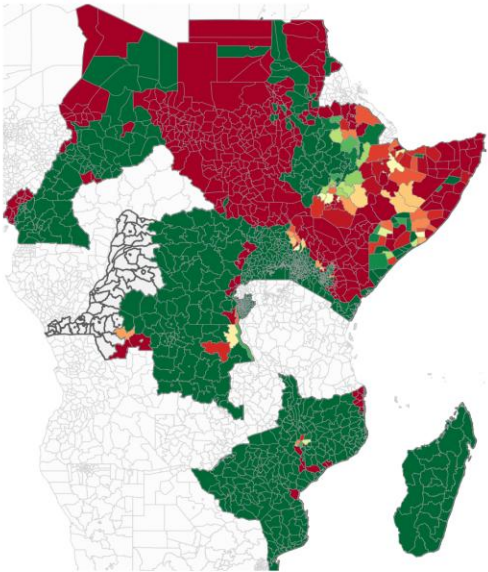
- Rather than relying on a single model, we propose an ensemble of models
- Vary how much the model focuses on transitions into crisis
 - Controlled through the weighting scale parameter
 - Low scale → reflects overall patterns, but may downplay rare transitions
 - High scale → amplifies transitions, highlighting emerging crises
- From conservative estimates -- To transition-sensitive signals
- For analysts, a set of informed signals to support interpretation:
 - Multiple complementary views of risk
 - Insight into where transitions may be under- or over-estimated
 - A way to triangulate signals across models

Near Term Projections (4 months ahead)

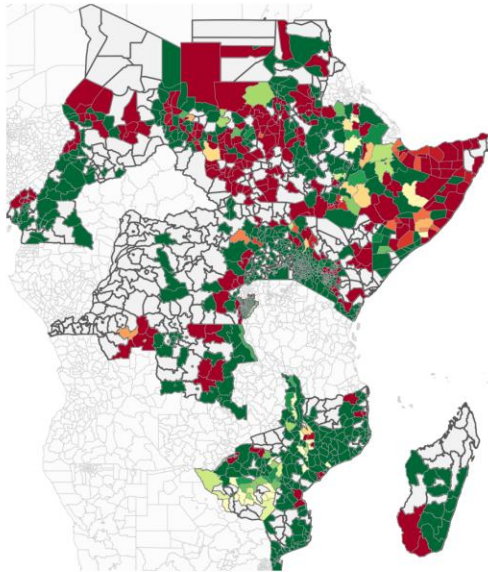
IPC3+ population per GAUL L2 (6/2023)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.117	0.297 (49%)	-0.097 (51%)	-
w = 100	0.113	0.232 (35%)	-0.281 (65%)	-
ML1	0.155	0.455 (23%)	-0.388 (19%)	24%

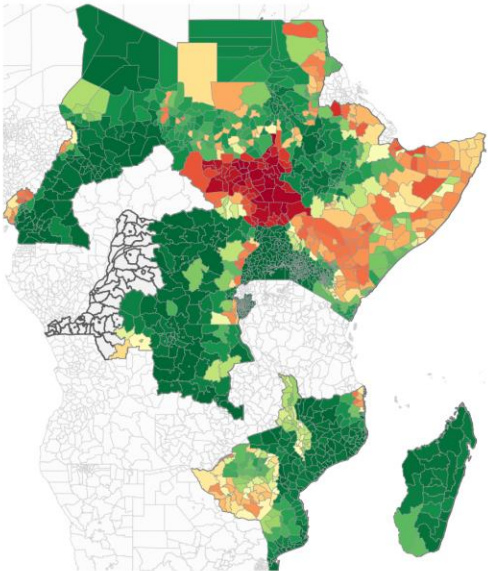
IPC (FEWSNET CS)



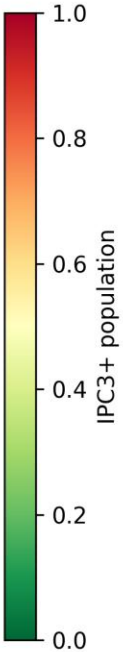
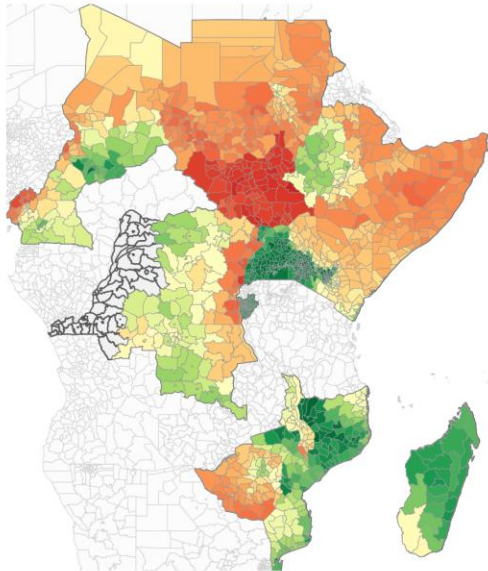
IPC (FEWSNET ML1)



IPC (Model (transition weight = 0))



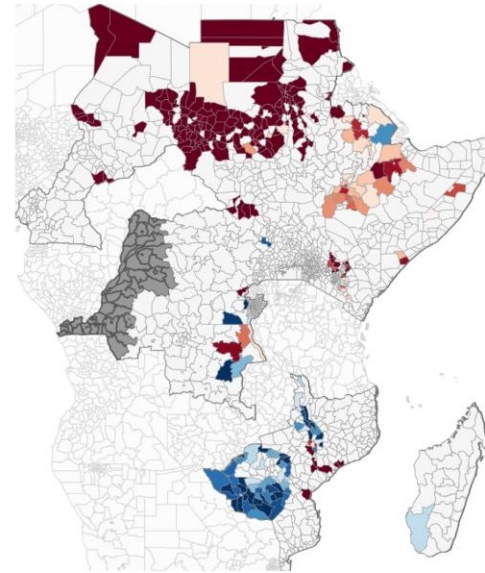
IPC (Model (transition weight = 100))



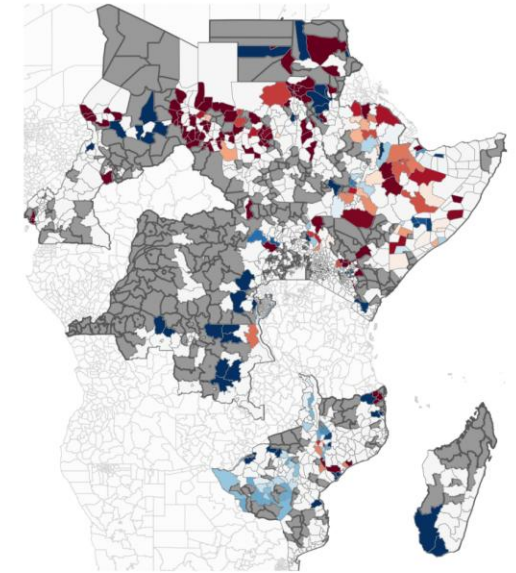
Observed change and model bias per GAUL L2 (2023-06)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.117	0.297 (49%)	-0.097 (51%)	-
w = 100	0.113	0.232 (35%)	-0.281 (65%)	-
ML1	0.155	0.455 (23%)	-0.388 (19%)	24%

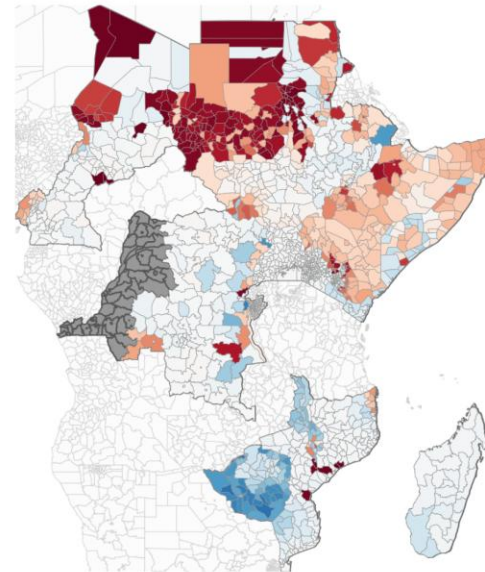
Observed change (y - y_last)



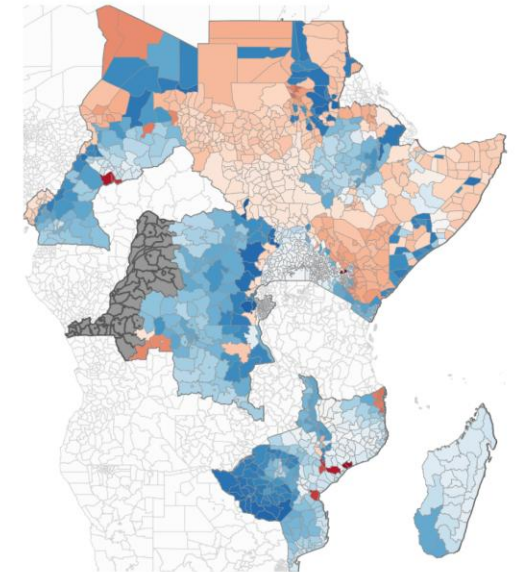
FEWSNET bias (prediction = ML1)



Model bias (transition weight = 0)



Model bias (transition weight = 100)



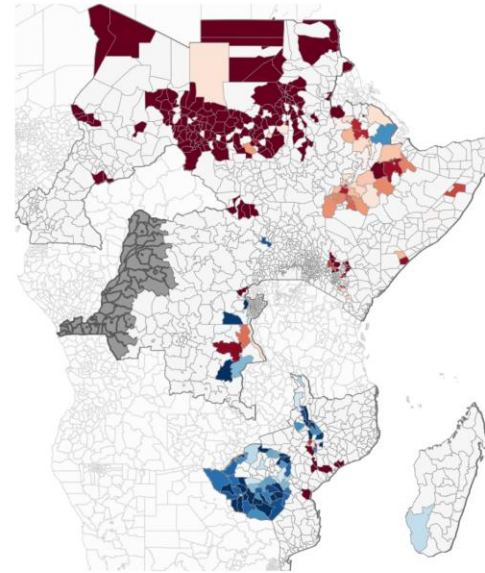
Signed difference



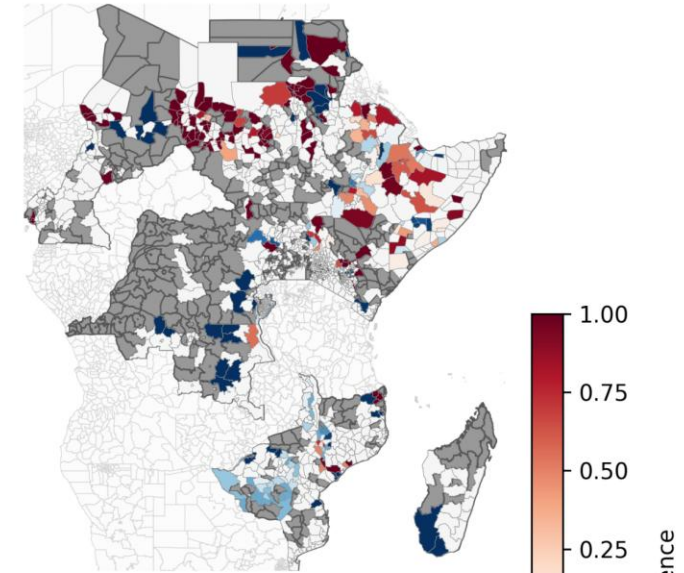
Observed change and model bias per GAUL L2 (2023-06)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.117	0.297 (49%)	-0.097 (51%)	-
w = 100	0.113	0.232 (35%)	-0.281 (65%)	-
ML1	0.155	0.455 (23%)	-0.388 (19%)	24%

Observed change (y - y_last)



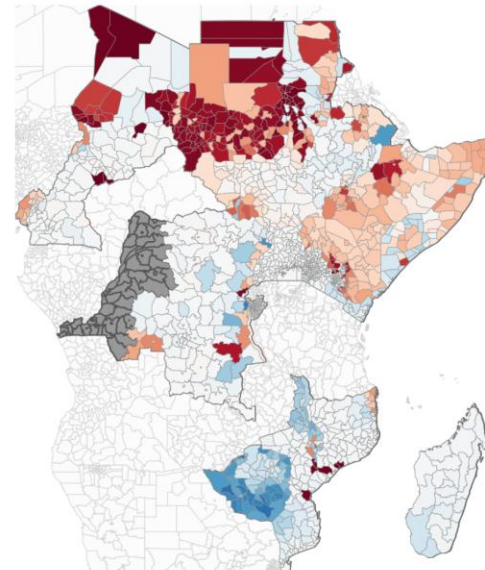
FEWSNET bias (prediction = ML1)



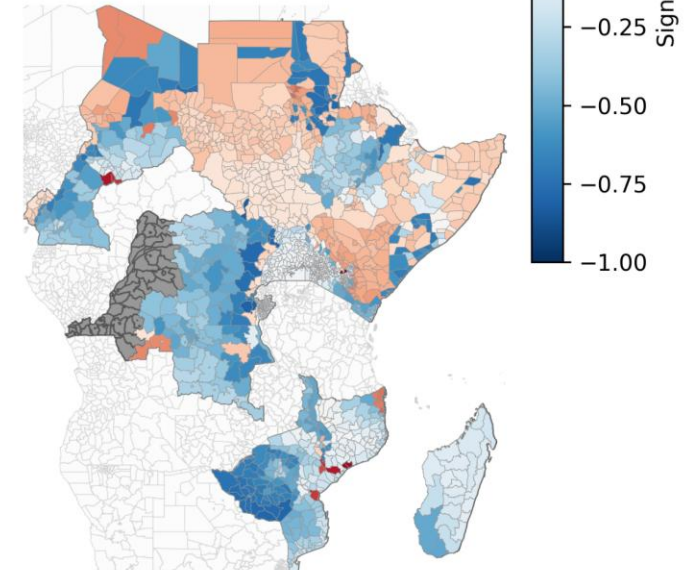
Binary classification results with a 0.5 cutoff:

Model	MSE	Type 2 (FN)	Type 1 (FP)	Correct
w = 0	0.157	1.000 (13%)	-1.000 (2%)	85%
w = 100	0.175	1.000 (2%)	-1.000 (16%)	82%
ML1	0.166	1.000 (10%)	-1.000 (6%)	64%

Model bias (transition weight = 0)

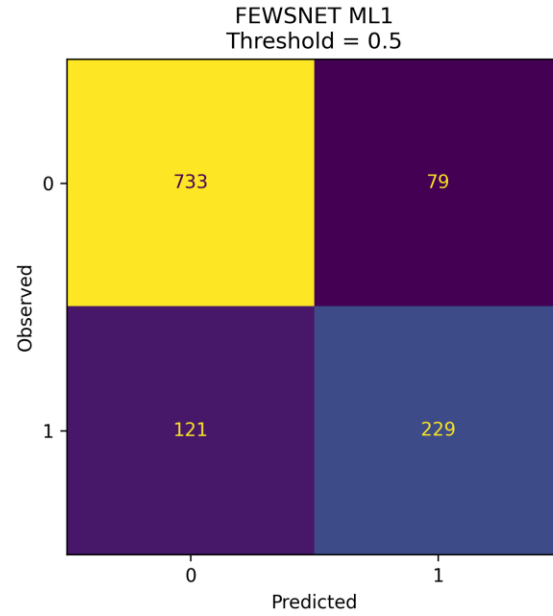


Model bias (transition weight = 100)

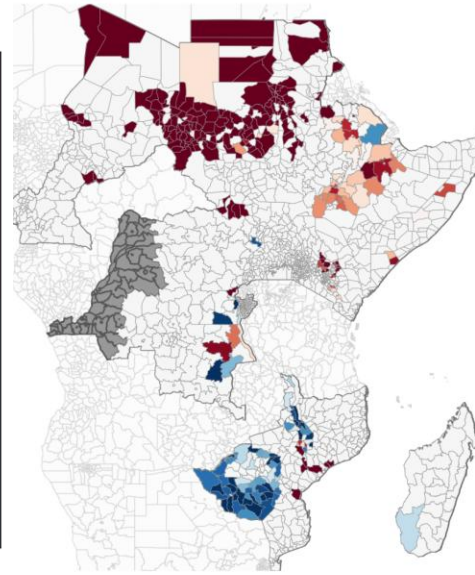


Observed change and model bias per GAUL L2 (2023-06)

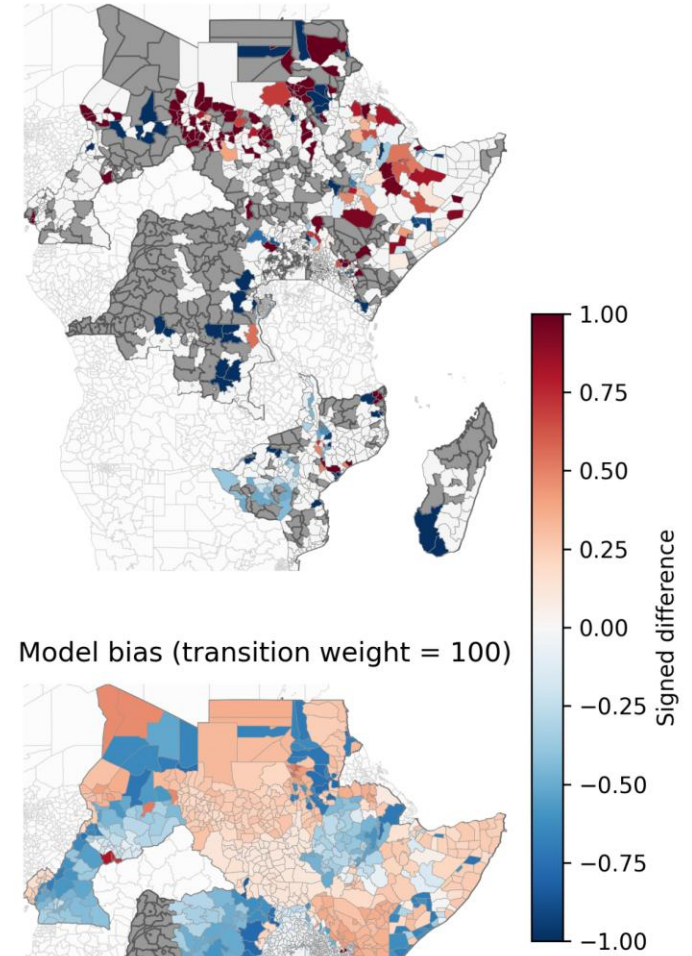
Binary diagnostics per GAUL L2 (6/2023)



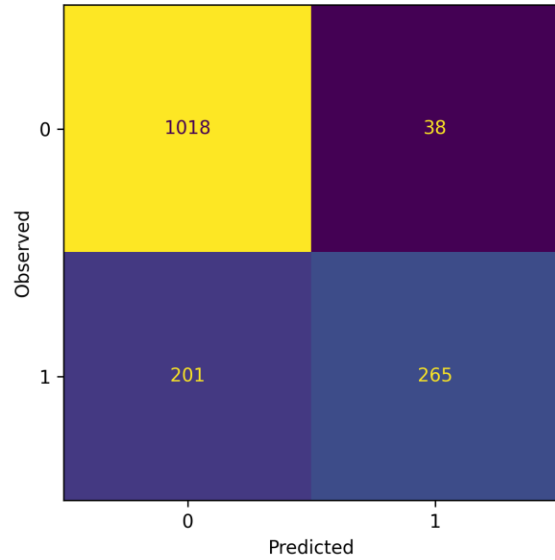
Observed change ($y - y_{\text{last}}$)



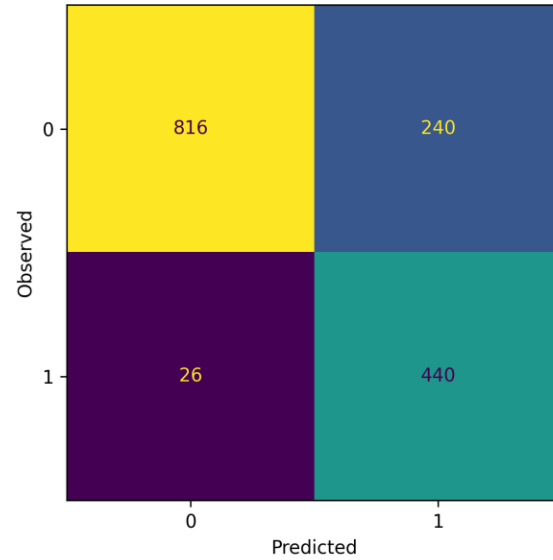
FEWSNET bias (prediction = ML1)



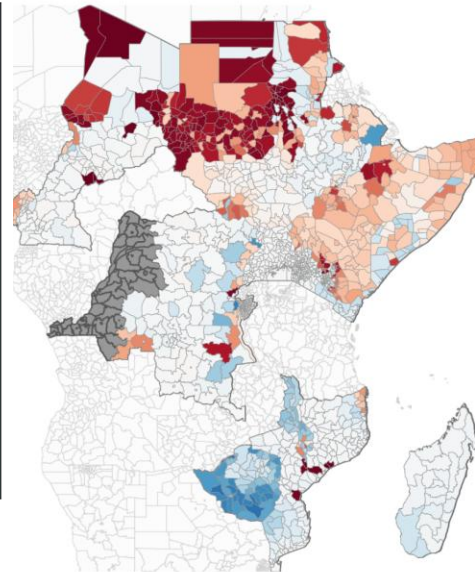
Model (transition weight = 0) Threshold = 0.5



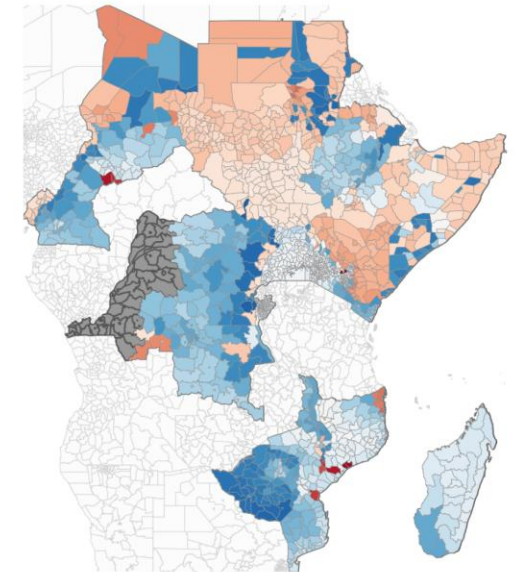
Model (transition weight = 100) Threshold = 0.5



Model bias (transition weight = 0)



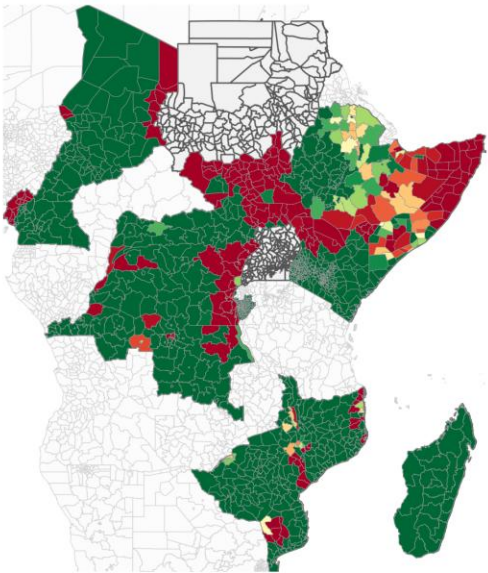
Model bias (transition weight = 100)



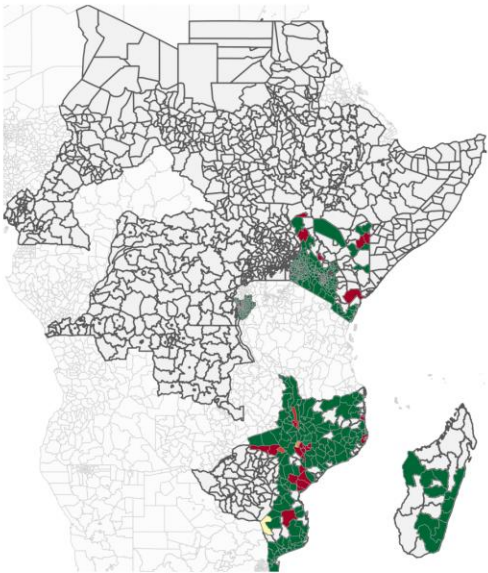
IPC3+ population per GAUL L2 (10/2025)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.074	0.245 (23%)	-0.144 (77%)	-
w = 100	0.161	0.166 (22%)	-0.366 (78%)	-
ML1	0.048	0.311 (8%)	-0.366 (7%)	59%

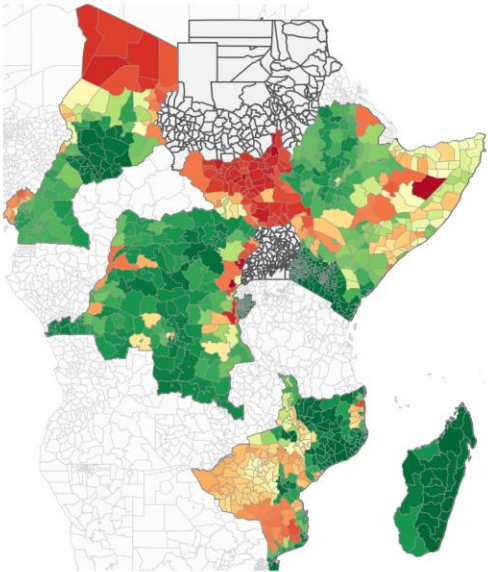
IPC (FEWSNET CS)



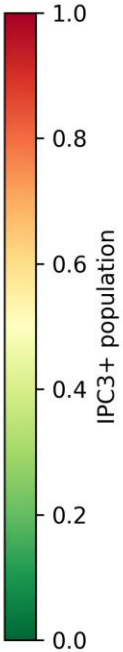
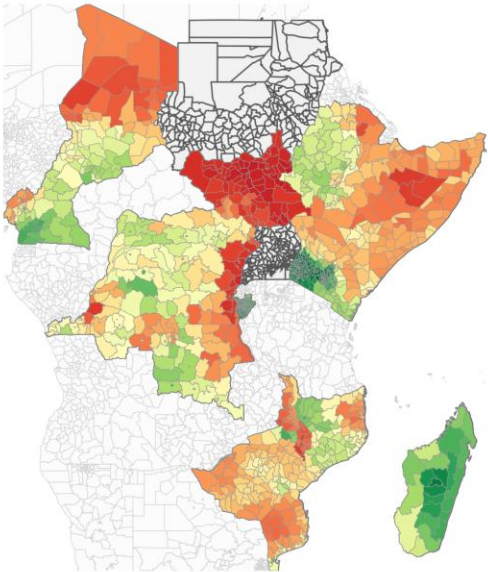
IPC (FEWSNET ML1)



IPC (Model (transition weight = 0))

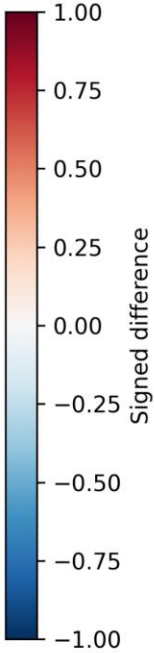
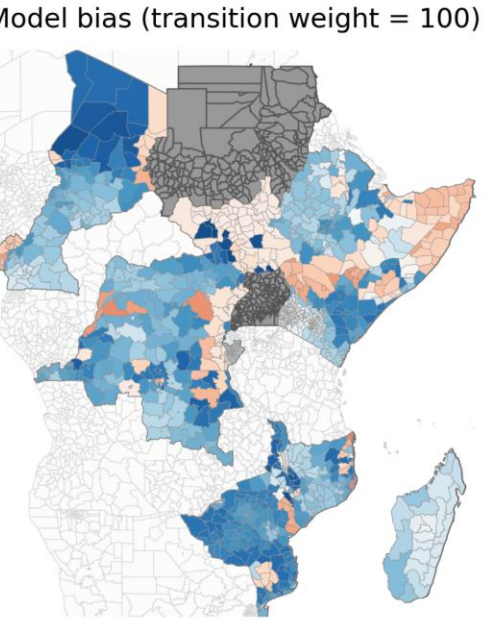
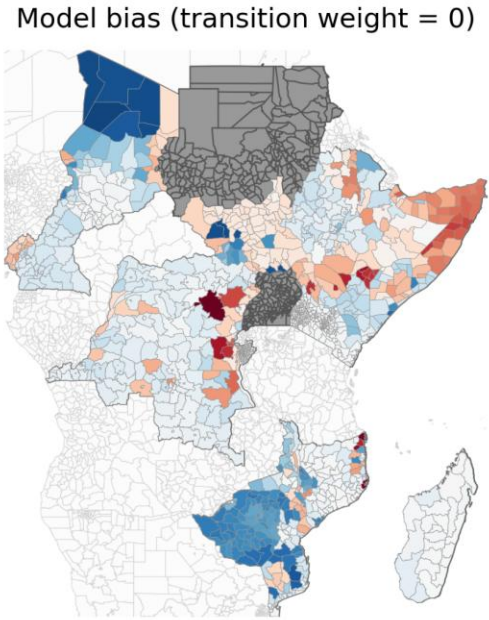
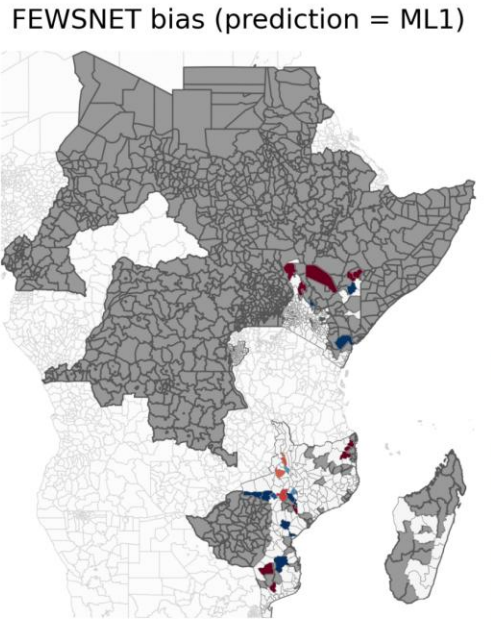
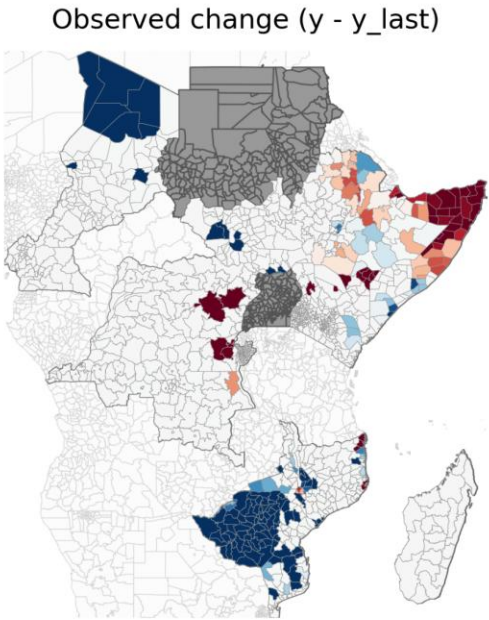


IPC (Model (transition weight = 100))



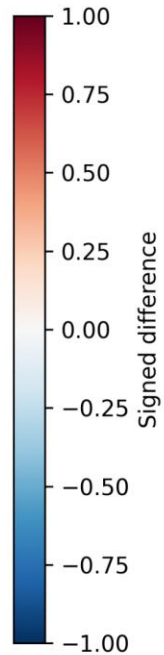
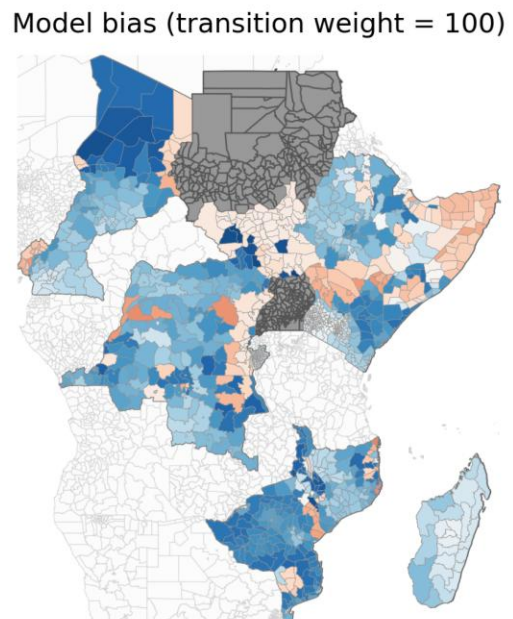
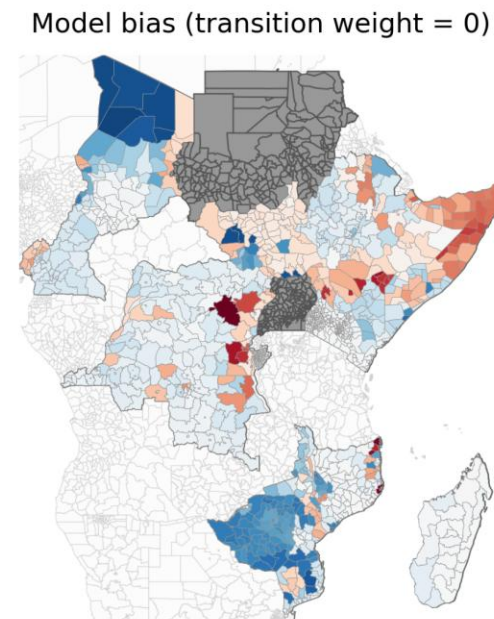
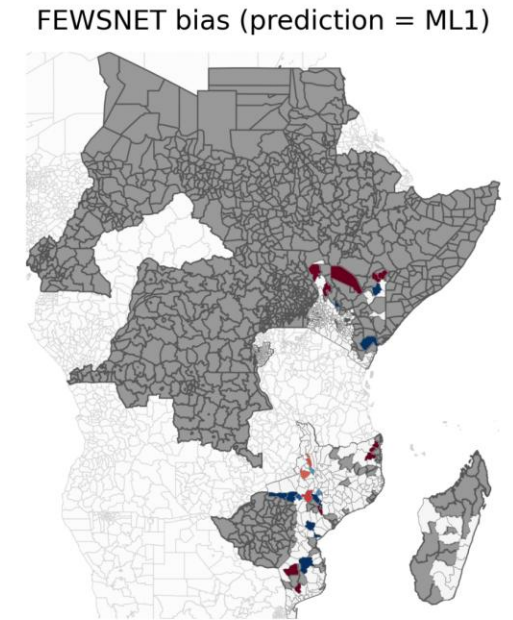
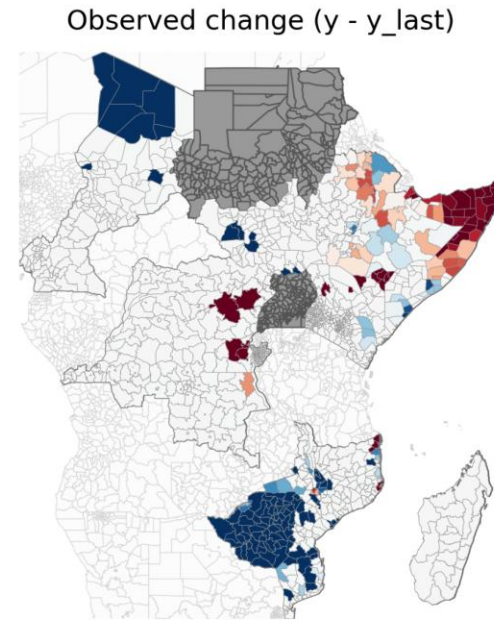
Observed change and model bias per GAUL L2 (2025-10)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.074	0.245 (23%)	-0.144 (77%)	-
w = 100	0.161	0.166 (22%)	-0.366 (78%)	-
ML1	0.048	0.311 (8%)	-0.366 (7%)	59%



Observed change and model bias per GAUL L2 (2025-10)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.074	0.245 (23%)	-0.144 (77%)	-
w = 100	0.161	0.166 (22%)	-0.366 (78%)	-
ML1	0.048	0.311 (8%)	-0.366 (7%)	59%

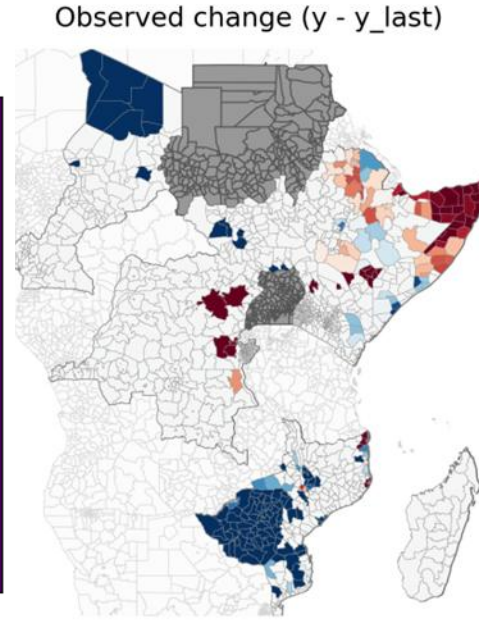
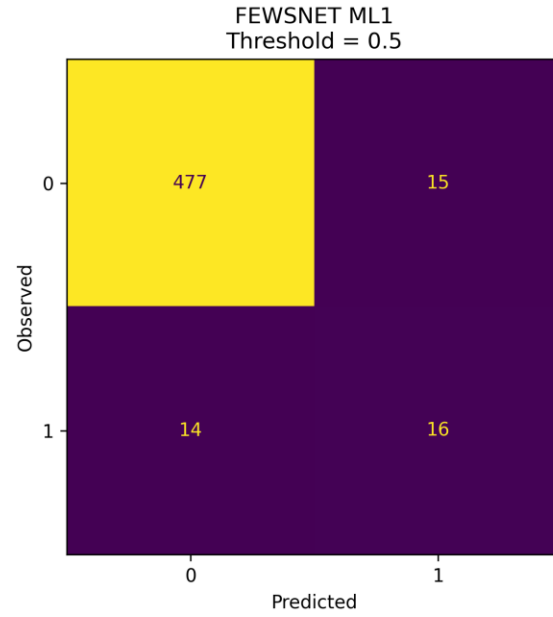


Binary classification results with a 0.5 cutoff:

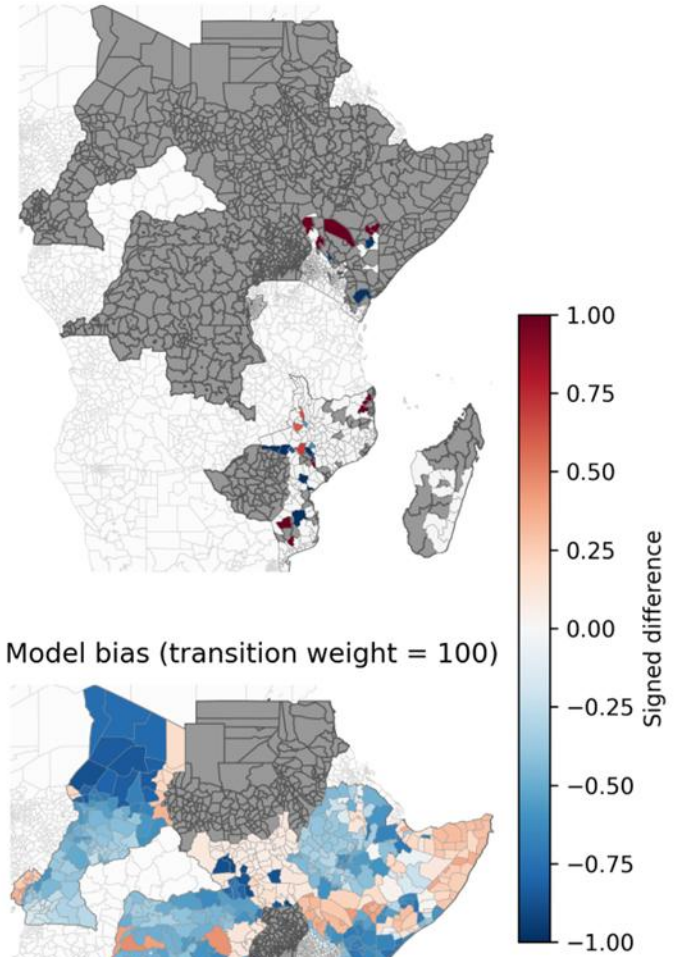
Model	MSE	Type 2 (FN)	Type 1 (FP)	Correct
w = 0	0.141	1.000 (4%)	-1.000 (10%)	86%
w = 100	0.271	1.000 (0%)	-1.000 (27%)	73%
ML1	0.056	1.000 (3%)	-1.000 (3%)	39%

Observed change and model bias per GAUL L2 (2025-10)

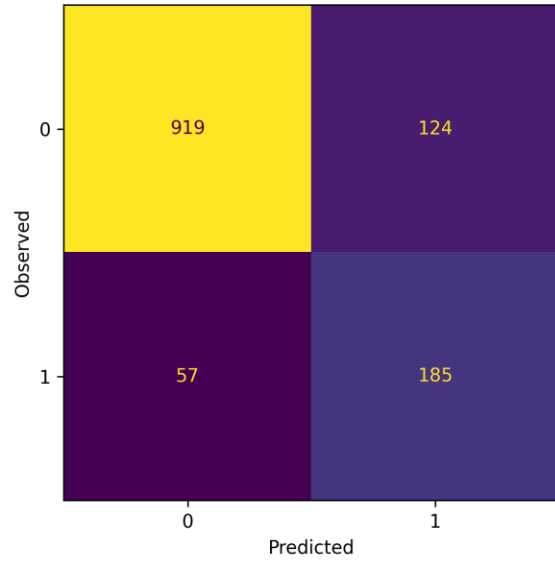
Binary diagnostics per GAUL L2 (10/2025)



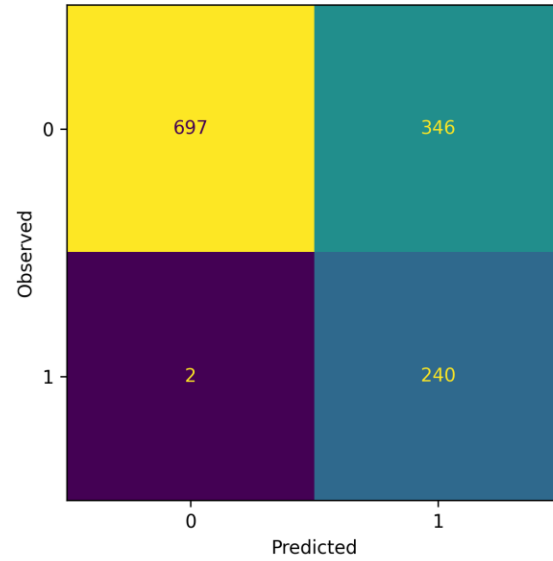
FEWSNET bias (prediction = ML1)



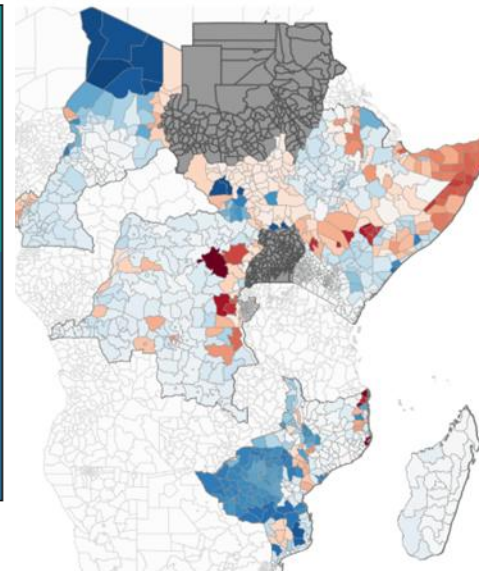
Model (transition weight = 0)
Threshold = 0.5



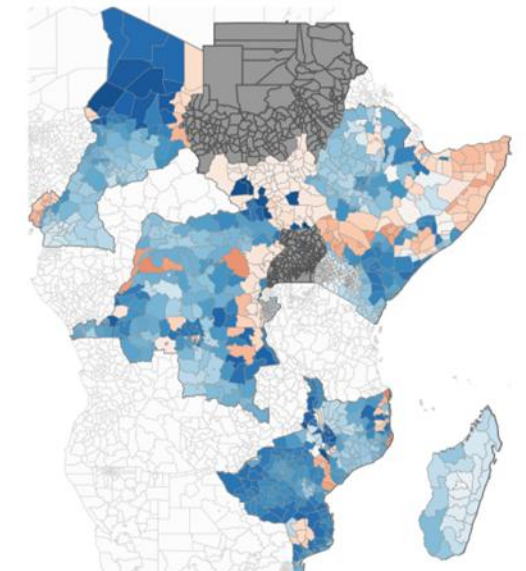
Model (transition weight = 100)
Threshold = 0.5



Model bias (transition weight = 0)



Model bias (transition weight = 100)

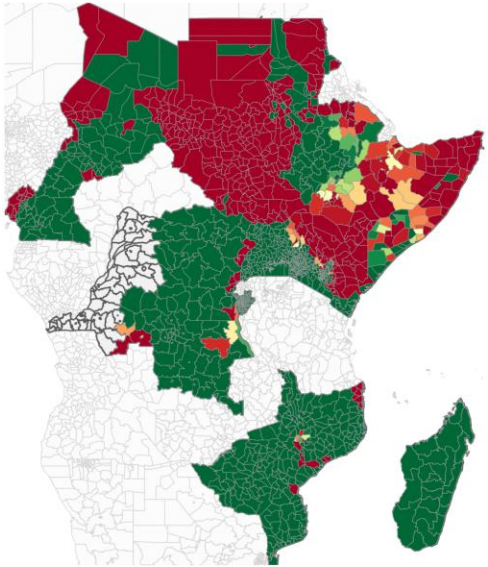


Current Situation Estimation

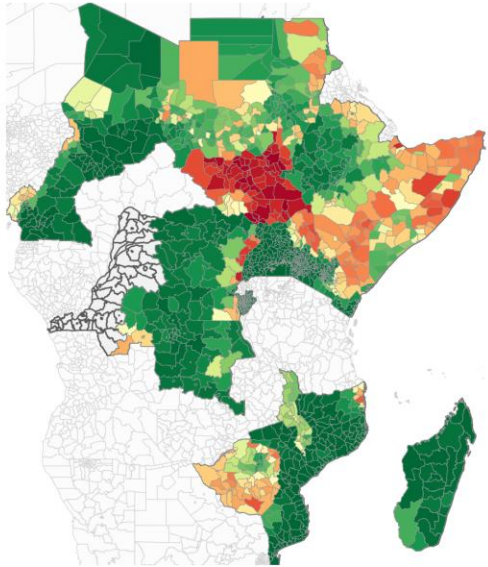
IPC3+ population per GAUL L2 (6/2023)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.113	0.347 (41%)	-0.084 (59%)	-
w = 100	0.123	0.219 (33%)	-0.304 (67%)	-
w = 10	0.093	0.250 (51%)	-0.146 (49%)	-

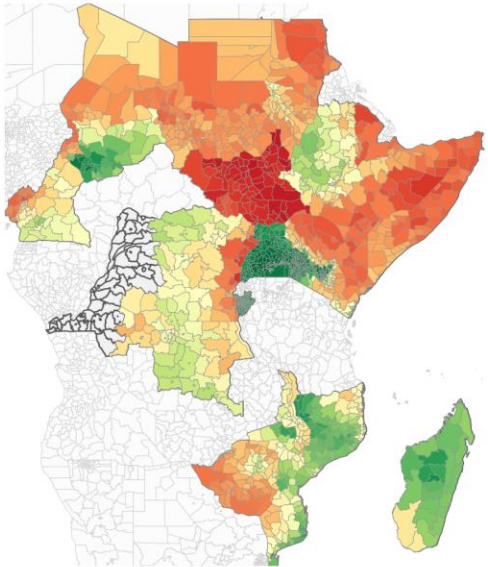
IPC (FEWSNET CS)



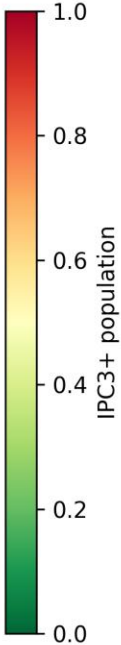
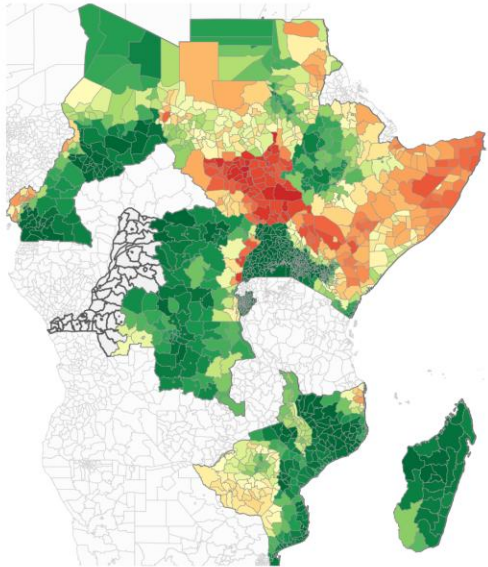
IPC (Model (transition weight = 0))



IPC (Model (transition weight = 100))



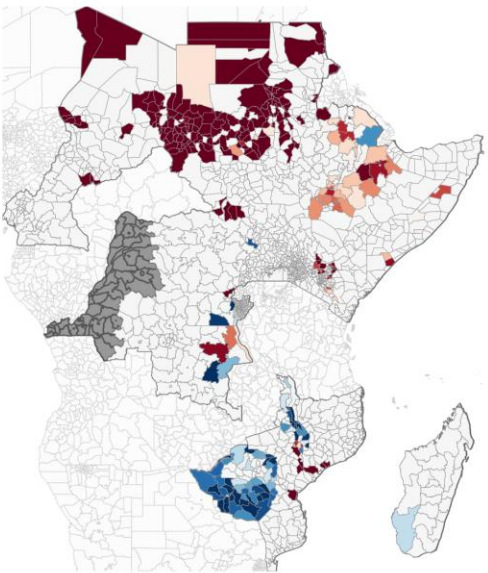
IPC (Model (transition weight = 10))



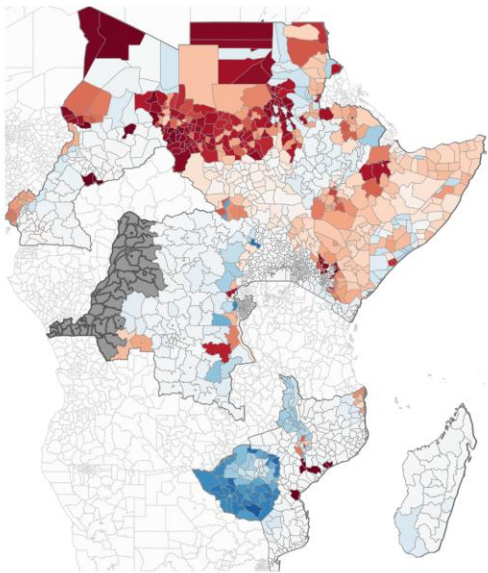
Observed change and model bias per GAUL L2 (2023-06)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.113	0.347 (41%)	-0.084 (59%)	-
w = 100	0.123	0.219 (33%)	-0.304 (67%)	-
w = 10	0.093	0.250 (51%)	-0.146 (49%)	-

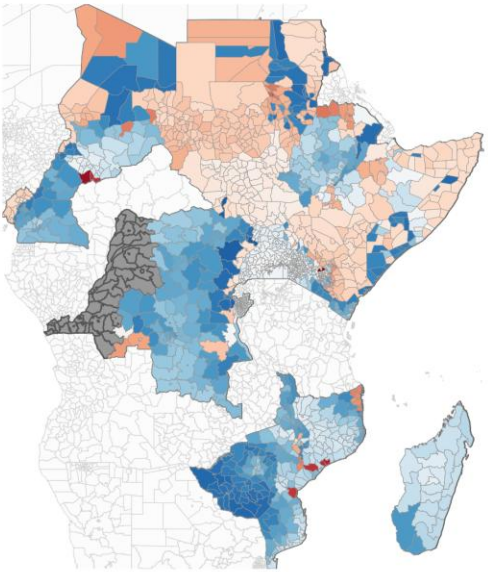
Observed change (y - y_last)



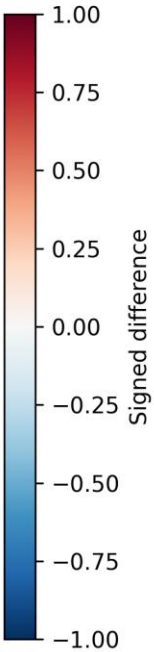
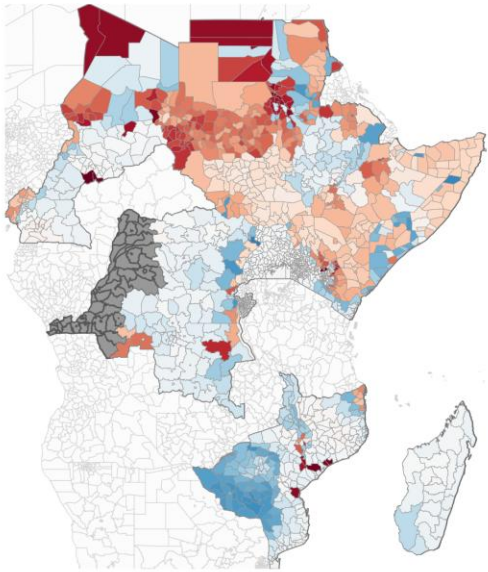
Model bias (transition weight = 0)



Model bias (transition weight = 100)



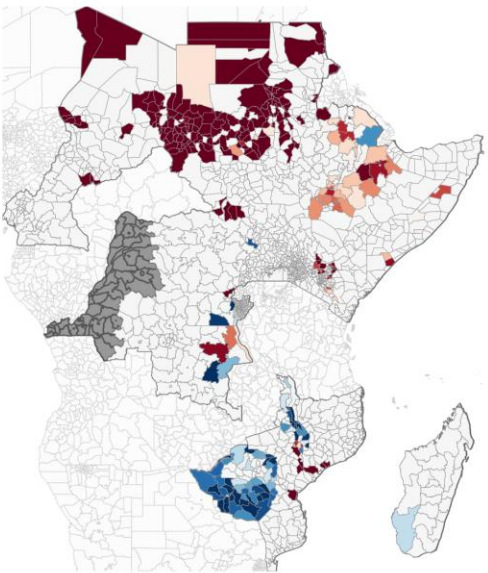
Model bias (transition weight = 10)



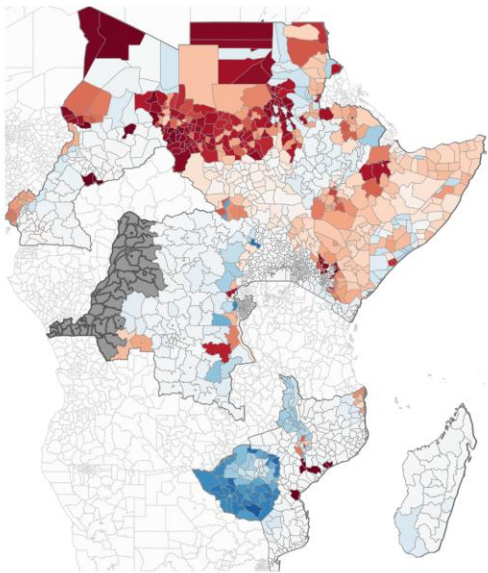
Observed change and model bias per GAUL L2 (2023-06)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.113	0.347 (41%)	-0.084 (59%)	-
w = 100	0.123	0.219 (33%)	-0.304 (67%)	-
w = 10	0.093	0.250 (51%)	-0.146 (49%)	-

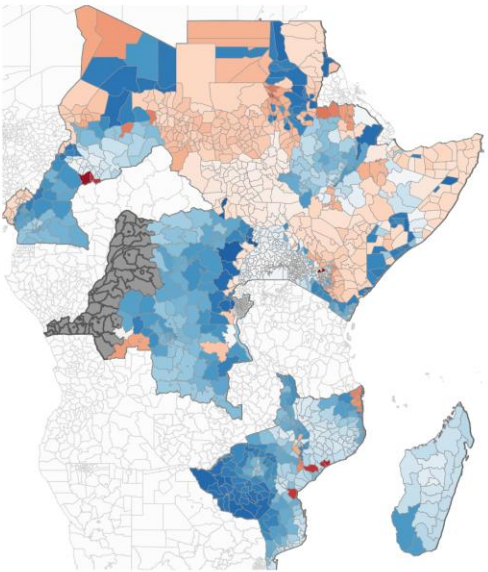
Observed change (y - y_last)



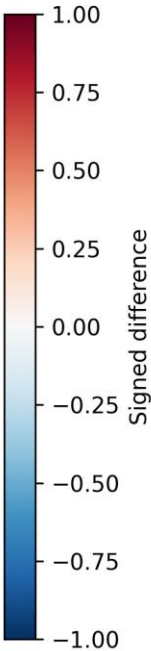
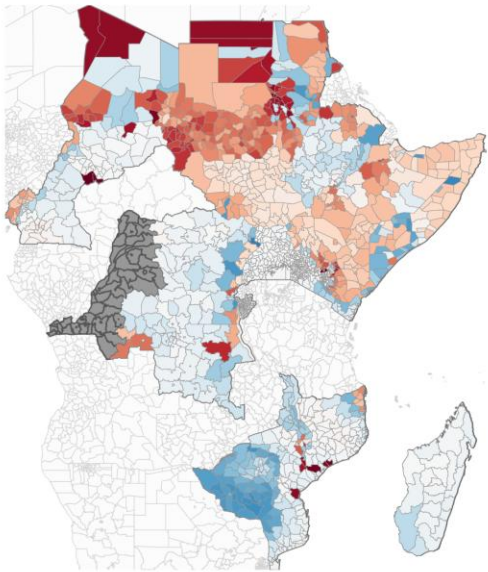
Model bias (transition weight = 0)



Model bias (transition weight = 100)



Model bias (transition weight = 10)

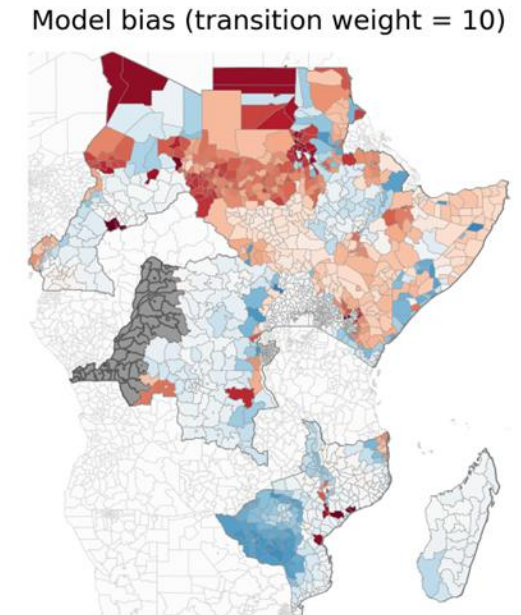
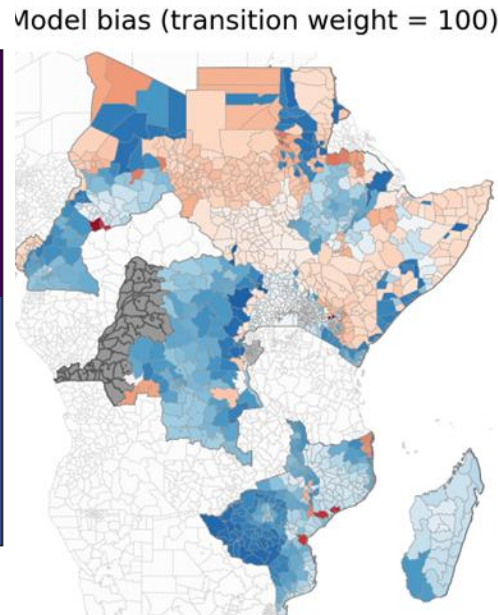
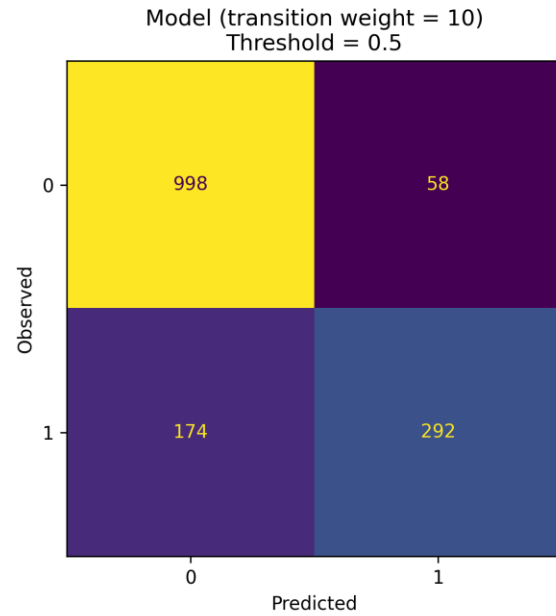
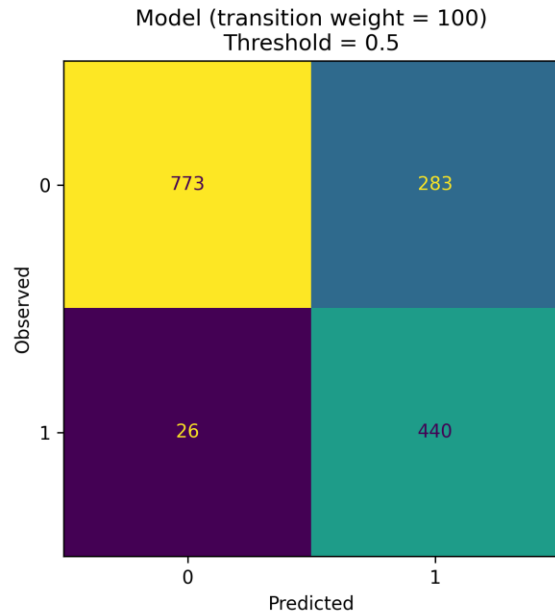
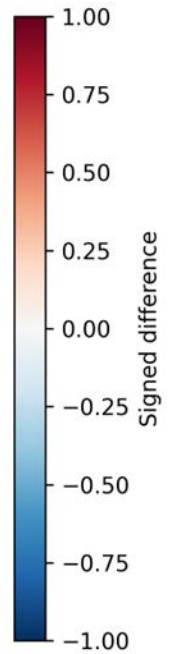
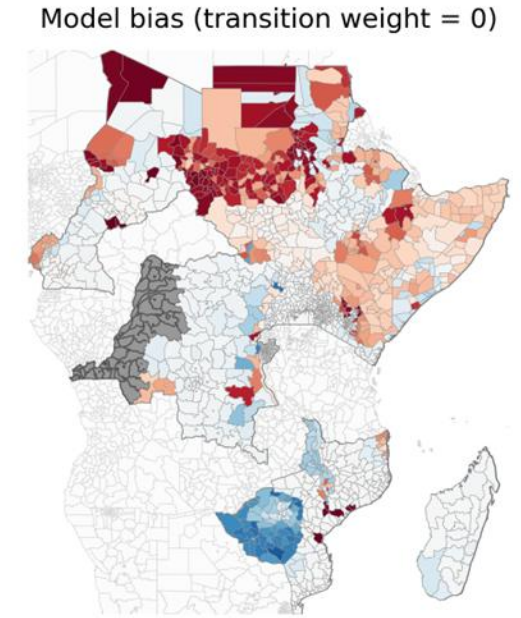
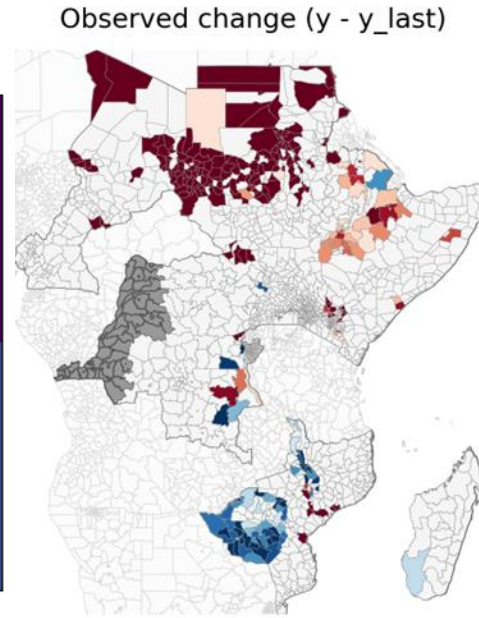
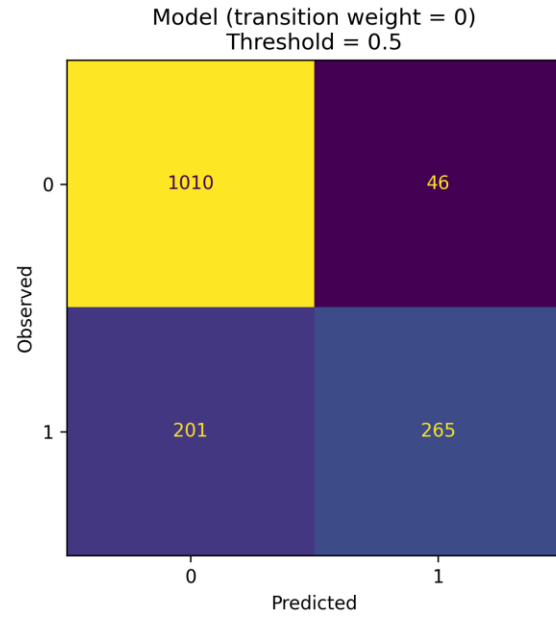


Binary classification results with a 0.5 cutoff:

Model	MSE	Type 2 (FN)	Type 1 (FP)	Correct
w = 0	0.162	1.000 (13%)	-1.000 (3%)	84%
w = 100	0.203	1.000 (2%)	-1.000 (19%)	79%
w = 10	0.152	1.000 (11%)	-1.000 (4%)	85%

Observed change and model bias per GAUL L2 (2023-06)

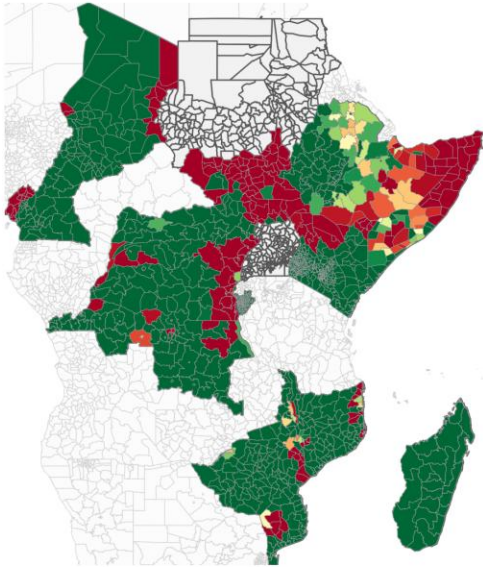
Binary diagnostics per GAUL L2 (6/2023)



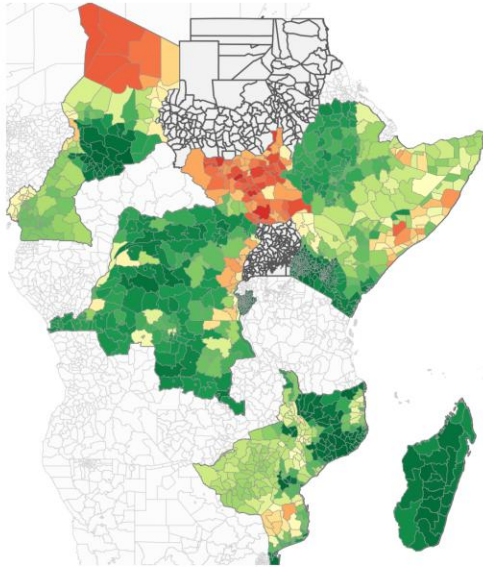
IPC3+ population per GAUL L2 (10/2025)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.069	0.338 (24%)	-0.125 (76%)	-
w = 100	0.144	0.141 (28%)	-0.153 (77%)	-
w = 10	0.066	0.263 (23%)	-0.130 (69%)	-

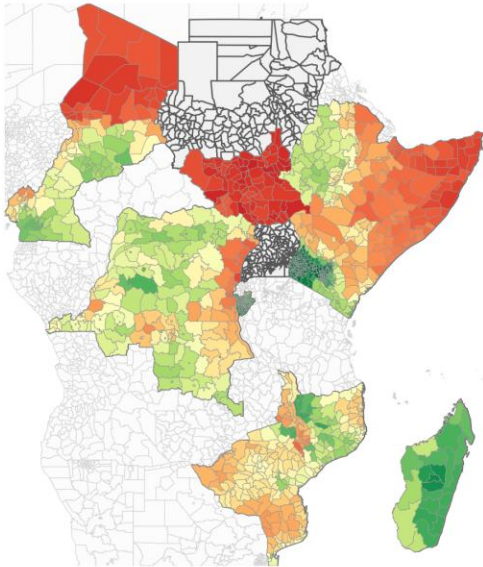
IPC (FEWSNET CS)



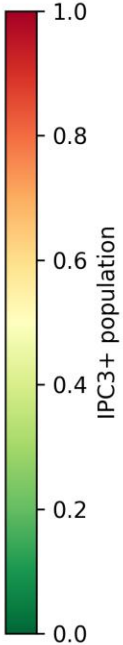
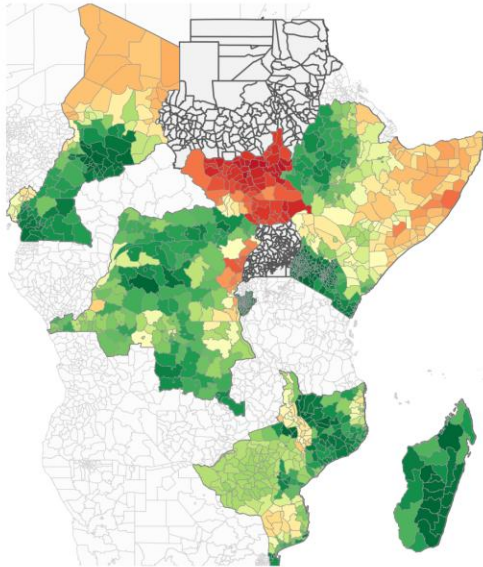
IPC (Model (transition weight = 0))



IPC (Model (transition weight = 100))



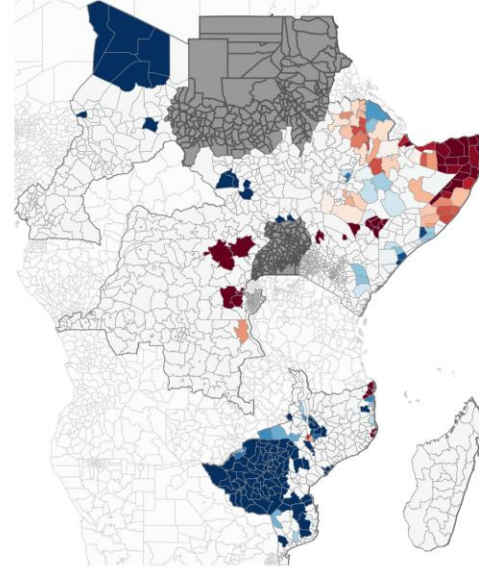
IPC (Model (transition weight = 10))



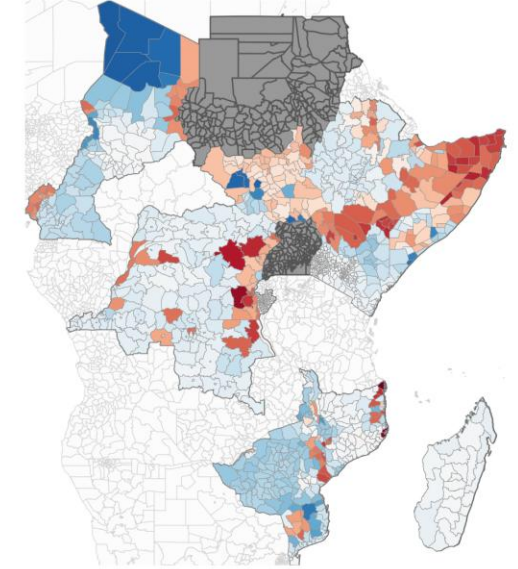
Observed change and model bias per GAUL L2 (2025-10)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.069	0.338 (24%)	-0.125 (76%)	-
w = 100	0.144	0.141 (28%)	-0.153 (77%)	-
w = 10	0.066	0.263 (23%)	-0.130 (69%)	-

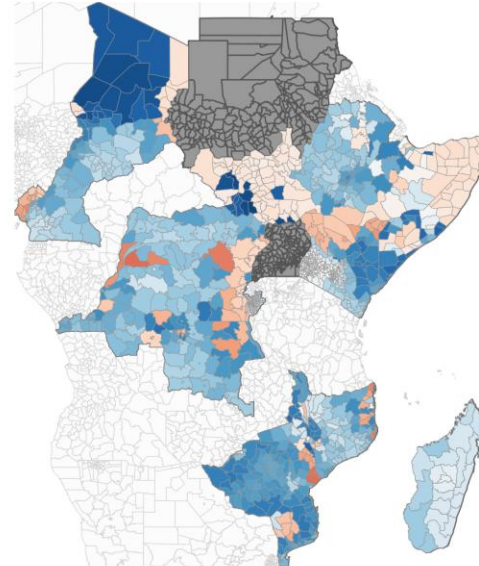
Observed change (y - y_last)



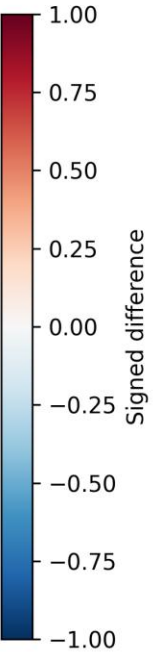
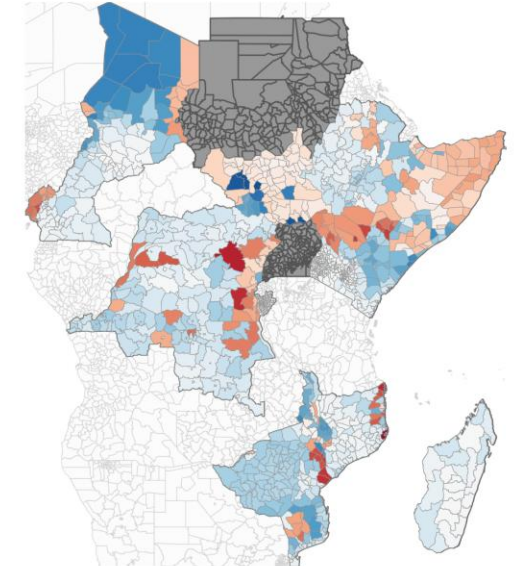
Model bias (transition weight = 0)



Model bias (transition weight = 100)



Model bias (transition weight = 10)



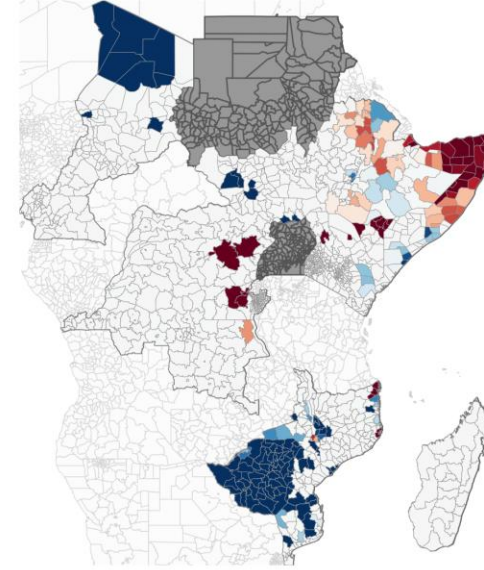
Observed change and model bias per GAUL L2 (2025-10)

Model	MSE	Type 2 (FN)	Type 1 (FP)	NA
w = 0	0.069	0.338 (24%)	-0.125 (76%)	-
w = 100	0.144	0.141 (28%)	-0.153 (77%)	-
w = 10	0.066	0.263 (23%)	-0.130 (69%)	-

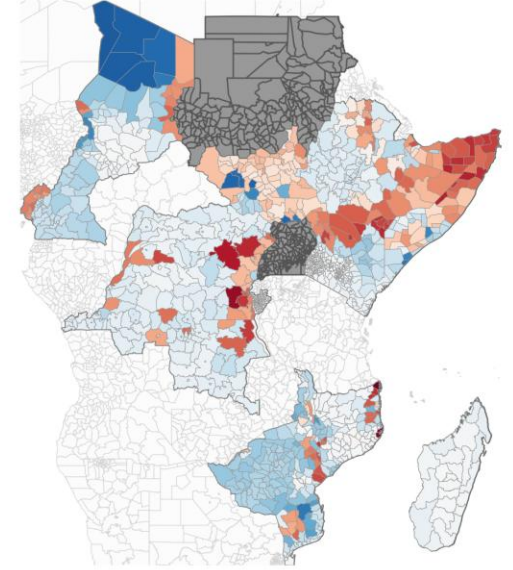
Binary classification results with a 0.5 cutoff:

Model	MSE	Type 2 (FN)	Type 1 (FP)	Correct
w = 0	0.111	1.000 (8%)	-1.000 (3%)	89%
w = 100	0.244	1.000 (1%)	-1.000 (24%)	75%
w = 10	0.104	1.000 (5%)	-1.000 (6%)	89%

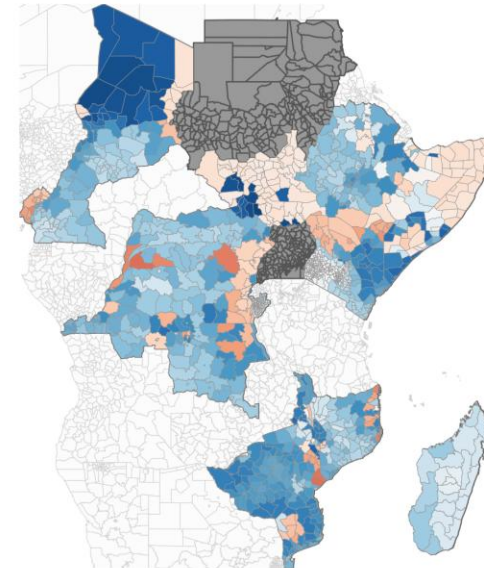
Observed change (y - y_last)



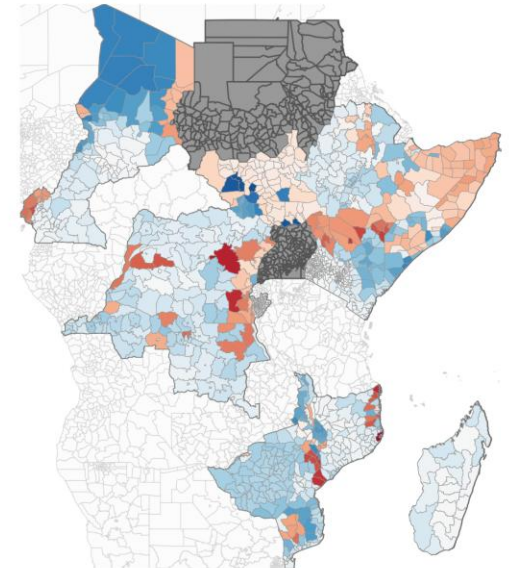
Model bias (transition weight = 0)



Model bias (transition weight = 100)



Model bias (transition weight = 10)



Signed difference



Observed change and model bias per GAUL L2 (2025-10)

Binary diagnostics per GAUL L2 (10/2025)

